

Trilingual Software Glossary / Glossaire Trilingue / المسرد الثلاثي اللغات

OCP Maintenance AI MVP -- Reliability Centered Maintenance Platform for Phosphate Processing

Audience	Language
Business stakeholders, management	Francais (primary)
Official documentation, regulatory	العربية (Arabic)
Technical staff, software engineers	English

Convention: Where a French or Arabic acronym differs from the English original, both are shown. Acronyms that are used identically across all three languages (e.g., SAP, OEE, RPN) are not translated.

Domain 1: Maintenance Methodology / Methodologie de

منهجية الصيانة / maintenance

English	Francais	العربية	Definition
RCM (Reliability Centered Maintenance)	MBF (Maintenance Basee sur la Fiabilite)	الصيانة المرتكزة على الموثوقية (MBF/RCM)	Systematic methodology for identifying the most effective maintenance strategy for each failure mode based on consequences and feasibility.
FMEA (Failure Mode and Effects Analysis)	AMDE (Analyse des Modes de Defaillance et de leurs Effets)	تحليل انماط الفشل وآثارها (AMDE)	Structured approach to identify potential failure modes, their causes, and effects on system performance. Each failure mode is defined as What + Mechanism + Cause.
FMECA (Failure Mode, Effects and Criticality Analysis)	AMDEC (Analyse des Modes de Defaillance, de leurs Effets et de leur Criticite)	تحليل انماط الفشل وآثارها وخرجيتها (AMDEC)	Extension of FMEA that adds criticality ranking via RPN (Severity x Occurrence x Detection) to prioritize risk mitigation actions.
CBM (Condition Based Maintenance)	MBC (Maintenance Basee sur la Condition)	الصيانة المبنية على الحالة	Maintenance strategy triggered by detectable equipment condition indicators (vibration, temperature, oil analysis) rather than fixed schedules. Requires a measurable P-F interval.
FT (Fixed Time Maintenance)	MT (Maintenance a Temps Fixe)	الصيانة بوقت ثابت	Scheduled restoration or discard at predetermined intervals. Applicable only to age-related failure patterns (A, B, C).
FFI (Failure Finding Interval)	IDD (Intervalle de Detection de Defaillance)	فترة اكتشاف الفشل	Periodic functional test to discover hidden failures in protective devices (e.g., safety relief valves, fire suppression systems).
RTF (Run to Failure)	MJD (Maintenance Jusqu'a la Defaillance)	التشغيل حتى الفشل	Deliberate decision to allow non-critical equipment to fail, applied when cost of prevention exceeds cost of failure and no safety/environmental consequences exist.
RPN (Risk Priority Number)	NPR (Nombre de Priorite de Risque)	رقم اولوية المخاطر	Quantitative risk score: $RPN = Severity \times Occurrence \times Detection$. Range 1--1000. Categories: LOW (1--49), MEDIUM (50--99), HIGH (100--199), CRITICAL (200--1000).
PDCA (Plan-Do-Check-Act)	PDCA (Planifier-Deployer-Contrôler-Agir)	خطط-نفذ-تحقق-صحح	Iterative four-phase management cycle for continuous improvement of processes and products.
CAPA (Corrective and Preventive Action)	ACAP (Action Corrective et Action Preventive)	الاجراءات التصحيحية والوقائية	Systematic process to investigate root causes of nonconformities, implement

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			corrective actions, and establish preventive measures.
VED Analysis (Vital-Essential-Desirable)	Analyse VED (Vital-Essentiel-Desirable)	VED تحليل حيوي-اساسي-مرغوب	Spare parts classification by operational criticality: Vital (production stops), Essential (degraded performance), Desirable (convenience).
FSN Analysis (Fast-Slow-Non-moving)	Analyse FSN (Rapide-Lent-Non-mobile)	FSN تحليل سريع-بطيء-راكد	Inventory classification by consumption velocity to optimize stock levels and identify obsolete materials.
ABC Analysis	Analyse ABC	ABC تحليل	Pareto-based inventory classification by annual consumption value: A (top 70--80%), B (15--20%), C (5--10%).
5W2H / 5W+2H	5W2H (Quoi-Quand-Ou-Qui-Pourquoi-Comment-Combien)	5W2H (ماذا-متى-اين-من-لماذا-كيف-كم)	Structured questioning framework: What, When, Where, Who, Why, How, How Much. Used for Level 1 quick RCA per GFSN REF-15.
Ishikawa Diagram (Fishbone / Cause-Effect)	Diagramme d'Ishikawa (Arete de poisson / Cause-Effet)	مخطط إيشيكawa (عظمة السمكة / السبب والاثـر)	Visual root cause analysis tool organizing potential causes into categories (Man, Machine, Method, Material, Milieu, Measurement).
Weibull Analysis	Analyse de Weibull	تحليل ويبـل	Statistical method using the Weibull distribution to model failure data. Beta (shape) identifies failure pattern; Eta (scale) estimates characteristic life.
Jack-Knife Analysis	Analyse Jack-Knife (Diagramme Jack-Knife)	Jack-Knife تحليل	Bad actor identification method plotting frequency vs. downtime to classify equipment as Acute, Chronic, Complex, or Controlled.
Pareto Analysis	Analyse de Pareto	تحليل باريتو	Prioritization technique based on the 80/20 principle: identifies the vital few contributors to the majority of failures or costs.
LCC (Life Cycle Cost)	CCV (Cout du Cycle de Vie)	تكلفة دورة الحياة	Total cost of ownership from acquisition through operation, maintenance, and disposal. Used to compare maintenance strategy economics.
OCR (Optimal Cost Replacement)	RCO (Remplacement a Cout Optimal)	الاستبدال بالتكلفة المثلى	Analysis to find the optimal preventive maintenance interval that minimizes total cost (PM cost + expected failure cost).

English	Francais	العربية	Definition
MoC (Management of Change)	GdC (Gestion du Changement)	إدارة التغيير	Formal process to evaluate, approve, implement, and close changes to equipment, procedures, or processes. Workflow: DRAFT - SUBMITTED - REVIEWING - APPROVED - IMPLEMENTING - CLOSED.
RBI (Risk-Based Inspection)	IBR (Inspection Basee sur le Risque)	التفتيش المبني على المخاطر	Methodology using a 5x5 probability-consequence risk matrix to prioritize inspection activities for static equipment.
P-F Interval	Intervalle P-F	فترة P-F	Time between the detectable onset of a potential failure (P) and functional failure (F). Determines CBM inspection frequency (inspect at intervals less than P-F).
MTBF (Mean Time Between Failures)	MTBF (Temps Moyen entre Defaillances)	متوسط الوقت بين الاعطال	Average elapsed time between consecutive failures of a repairable system. Key reliability KPI.
MTTR (Mean Time to Repair)	MTTR (Temps Moyen de Reparation)	متوسط وقت الإصلاح	Average time required to restore a failed system to operational status. Key maintainability KPI.
OEE (Overall Equipment Effectiveness)	TRG (Taux de Rendement Global)	الفعالية الشاملة للمعدات	Composite metric: $OEE = Availability \times Performance \times Quality$. Measures manufacturing productivity.
Asset Health Index	Indice de sante de l'actif	مؤشر صحة الاصل	Composite score reflecting current equipment condition based on criticality, backlog, failure mode coverage, and pending work.
Failure Pattern A (Bathtub)	Modele de defaillance A (Baignoire)	A نمط الفشل (حوض الاستحمام)	Nowlan and Heap pattern: high infant mortality, constant rate, then wear-out. Low prevalence in industrial equipment.
Failure Pattern B (Age-Related)	Modele de defaillance B (Lie a l'age)	B نمط الفشل (مرتبط بالعمر)	Increasing failure probability with age. Applicable to simple components subject to fatigue or corrosion.
Failure Pattern C (Fatigue)	Modele de defaillance C (Fatigue)	C نمط الفشل (الاجهاد)	Slowly increasing failure probability. No distinct wear-out region. Moderate prevalence.
Failure Pattern D (Stress-Related)	Modele de defaillance D (Lie au stress)	D نمط الفشل (مرتبط بالاجهاد)	Constant failure probability at any age after initial break-in. Moderate prevalence.

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Failure Pattern E (Random)	Modele de defaillance E (Aleatoire)	E نمط الفشل (عشوائي)	Constant failure probability at any age. High prevalence in complex systems.
Failure Pattern F (Early Life)	Modele de defaillance F (Mortalite infantile)	F نمط الفشل (الحياة المبكرة)	Highest failure rate when new, decreasing to constant. Most common pattern in industrial equipment.
Root Cause Analysis (RCA)	Analyse des Causes Racines (ACR)	تحليل السبب الجذري	Systematic investigation to identify the fundamental cause of a failure event. GFSN 3-level methodology: Physical - Human - Latent causes.
Defect Elimination (DE)	Elimination des Defaults (ED)	ازالة العيوب	GFSN 5-stage program: Identify - Prioritize - Analyze - Implement - Control. Aims to permanently remove recurring failure causes.
Backlog	Arriere (de maintenance)	تراكم (اعمال الصيانة)	Accumulated maintenance work requests and orders awaiting execution. Measured in total hours.
Work Order (WO)	Ordre de Travail (OT)	امر العمل	Formal authorization document specifying maintenance work to be performed, including resources, materials, and scheduling.
Maintenance Plan	Plan de maintenance	خطة الصيانة	Scheduled set of maintenance activities for an asset, defining tasks, frequencies, and resource requirements.
Task List	Gamme de maintenance	قائمة المهام	Ordered sequence of maintenance operations (inspect, check, test, lubricate, replace, repair) assigned to a work package.
Work Package (WP)	Paquet de Travail (PT)	حزمة العمل	Grouped set of maintenance tasks sharing the same labour type, constraint (online/offline), and frequency. Types: Standalone, Suppressive, Sequential.
Work Instruction	Instruction de Travail	تعليمات العمل	Detailed step-by-step procedure for executing a specific maintenance task, including safety precautions and quality criteria.
Criticality Assessment	Evaluation de la criticite	تقييم الحرجية	11-criteria risk assessment matrix (Safety, Health, Environment, Production, Operating Cost, Capital Cost, Schedule, Revenue, Communications, Compliance,

English	Francais	العربية	Definition
			Reputation) x 5 likelihood levels producing 4 risk classes (I--IV).
Hidden Failure	Defaillance cachee	فشل مخفي	Failure that is not evident to operating crew under normal conditions. Requires failure-finding tasks (FFI) to detect. Typically applies to protective devices.
Evident Failure	Defaillance evidente	فشل ظاهر	Failure that becomes apparent to operating crew during normal operations through observable symptoms.
Functional Failure	Defaillance fonctionnelle	الفشل الوظيفي	Inability of an asset to perform its required function at the required performance standard. Can be Total (complete loss) or Partial (degraded performance).
Failure Mode	Mode de defaillance	نمط الفشل	Specific way in which a component fails, defined as What (component) + Mechanism (how it fails) + Cause (why it fails). 72 valid mechanism-cause combinations per SRC-09.
Failure Effect	Effet de la defaillance	تأثير الفشل	Observable consequence of a failure mode: evidence, safety threat, production impact, collateral damage, and repair requirements.
Failure Consequence	Consequence de la defaillance	عاقبة الفشل	Classification of failure impact: Hidden Safety, Hidden Non-Safety, Evident Safety, Evident Environmental, Evident Operational, Evident Non-Operational.

Domain 2: OCP / Phosphate Industry / Industrie phosphatiere /

صناعة الفوسفات

English	Francais	العربية	Definition
Grinding	Broyage	الطحن	Size reduction of phosphate rock using mechanical force in mills (SAG, ball). Primary beneficiation step to liberate phosphate minerals from gangue.
Flotation	Flottation	التعويم	Separation process using air bubbles and chemical reagents to selectively concentrate phosphate minerals from ground ore in flotation cells.
Sedimentation	Sedimentation	الترسيب	Gravity-driven separation of solid particles from slurry in thickeners. Produces underflow (concentrated solids) and overflow (clarified water).
Filtration	Filtration	الترشيح	Mechanical separation of solids from liquids using belt filters or pressure filters to dewater phosphate concentrate.
Drying	Sechage	التجفيف	Thermal removal of moisture from phosphate concentrate using rotary dryers to achieve target moisture content for transport or processing.
Conveying	Convoyage	النقل بالسيور	Continuous transport of bulk materials (phosphate rock, concentrate, product) via belt conveyors between processing stages.
Storage	Stockage	التخزين	Bulk storage of raw materials, intermediates, and finished products in silos, stockpiles, or warehouses.
Pumping	Pompage	الضخ	Transfer of slurry, water, or reagents through pipelines using centrifugal or positive displacement pumps.
SAG Mill	Broyeur SAG (Broyeur semi-autogene)	شبه SAG طاحونة (ذاتية الطحن)	Semi-Autogenous Grinding mill using a combination of ore and steel balls as grinding media. Primary grinding stage in phosphate beneficiation.
Ball Mill	Broyeur a boulets	طاحونة كرات	Rotating cylindrical mill using steel balls as grinding media for secondary/fine grinding of phosphate ore.

English	Français	العربية	Definition
Slurry Pump	Pompe a boue (Pompe a pulpe)	مضخة الملاط	Heavy-duty centrifugal pump designed to transport abrasive slurry mixtures of ground phosphate rock and water.
Thickener	Epaississeur	المكثف	Large settling tank where flocculated slurry is concentrated by gravity. Produces thickened underflow and clear overflow water for recycling.
Flotation Cell	Cellule de flottation	خلية التعويم	Vessel in which air is injected into phosphate slurry with reagents; phosphate-bearing froth is collected from the surface.
Belt Conveyor	Convoyeur a bande	سير ناقل	Continuous loop belt system for transporting bulk phosphate materials between processing stages. Key components: pulleys, idlers, belt, drive.
Belt Filter	Filtre a bande	مرشح شريطي	Continuous vacuum or pressure filter using a moving belt to dewater phosphate slurry into filter cake.
Rotary Dryer	Secheur rotatif	مجفف دوار	Cylindrical rotating drum through which hot gas passes to evaporate moisture from wet phosphate concentrate.
Crusher	Concasseur	الكسارة	Machine that reduces large phosphate rock pieces to smaller sizes by compression or impact. Types: jaw, cone, gyratory, impact.
Screen / Classifier	Crible / Classificateur	غربال / مصنف	Equipment that separates particles by size using vibrating screens (dry) or hydrocyclones (wet classification).
Cyclone (Hydrocyclone)	Cyclone (Hydrocyclone)	سيكلون (هيدروسيكلون)	Centrifugal separator that classifies slurry particles by size and density. Overflow (fines) and underflow (coarse) are directed to different process stages.
JFC (Jorf Fertilizer Complex)	JFC (Complexe d'Engrais de Jorf Lasfar)	مجمع الاسمدة بالجرف الاصفر (JFC)	OCP's integrated phosphate processing and fertilizer production complex at Jorf Lasfar, Morocco.
Equipment Tag	TAG Equipement	علامة المعدات	Unique alphanumeric identifier for physical equipment following OCP naming convention (e.g., BRY-SAG-ML-001). Used for field identification and CMMS cross-reference.

English	Francais	العربية	Definition
Functional Location	Poste technique (Emplacement fonctionnel)	الموقع الوظيفي	Hierarchical position in the plant structure representing where maintenance is performed (SAP TPLNR). Maps physical plant topology.
Phosphate Rock	Roche phosphatee (Minerai de phosphate)	صخر الفوسفات	Raw mineral ore containing phosphorus, the primary feedstock for OCP operations. Extracted from Moroccan deposits and processed through beneficiation.
Beneficiation	Enrichissement (Valorisation)	الاثراء (التحسين)	Series of physical and chemical processes (crushing, grinding, flotation, drying) to increase the P2O5 content of raw phosphate rock to commercial grade.

Domain 3: SAP PM (Plant Maintenance) / SAP PM / SAP PM (صيانة)

English	Francais	العربية	Definition
PM01 (Inspection Order)	PM01 (Ordre d'inspection)	PM01 (امر الفحص)	SAP work order type for condition monitoring, inspection rounds, and diagnostic checks. Non-intrusive maintenance.
PM02 (Preventive Maintenance Order)	PM02 (Ordre de maintenance preventive)	PM02 (امر الصيانة الوقائية)	SAP work order type for scheduled preventive maintenance tasks: lubrication, replacement, overhaul at fixed intervals.
PM03 (Corrective Maintenance Order)	PM03 (Ordre de maintenance corrective)	PM03 (امر الصيانة التصحيحية)	SAP work order type for unplanned repairs and breakdown maintenance following a detected failure.
TPLNR (Technical Place Number)	TPLNR (Numero de poste technique)	TPLNR (رقم الموقع الفني)	SAP field identifier for Functional Location. Hierarchical key encoding plant structure (e.g., JFC1-MIN-BRY-01).
EQUNR (Equipment Number)	EQUNR (Numero d'equipement)	EQUNR (رقم المعدات)	SAP unique identifier for individual equipment items. Links to functional location, maintenance history, and BOM.
LOTO (Lock Out Tag Out)	LOTO (Consignation / Deconsignation)	LOTO (القفل والتعليق)	Safety procedure requiring isolation and tagging of energy sources before maintenance work. Mandatory for all offline tasks.
Work Center	Centre de travail (Poste de travail)	مركز العمل	SAP organizational unit representing a group of technicians with specific skills (e.g., Mechanical Workshop, Electrical Team). Defines available capacity.
Planner Group	Groupe de planificateurs	مجموعة المخططين	SAP organizational unit for maintenance planners responsible for a specific plant area or equipment category.
Task List (SAP)	Gamme (Liste de taches SAP)	قائمة المهام (SAP)	SAP object containing the ordered sequence of maintenance operations, materials, and time estimates for a work order.
Maintenance Plan (SAP)	Plan de maintenance (SAP)	خطة الصيانة (SAP)	SAP scheduling object that generates work orders automatically based on time or performance-based cycles. Links to maintenance items and task lists.

English	Francais	العربية	Definition
Maintenance Item	Poste de maintenance (Element de maintenance)	عنصر الصيانة	SAP object linking a maintenance plan to a specific piece of equipment or functional location, defining what is maintained.
System Condition 1 (Operating)	Etat systeme 1 (En fonctionnement)	حالة النظام 1 (تشغيل)	SAP system status indicating maintenance can be performed while equipment is running (online tasks).
System Condition 2 (Not Operating)	Etat systeme 2 (A l'arret)	حالة النظام 2 (متوقف)	SAP system status indicating equipment must be shut down for maintenance (offline tasks).
System Condition 3 (Test Mode)	Etat systeme 3 (Mode test)	حالة النظام 3 (وضع الاختبار)	SAP system status indicating equipment is in test or commissioning mode during maintenance.
Notification (SAP)	Avis de maintenance (Notification SAP)	اشعار الصيانة (SAP)	SAP document recording a maintenance-relevant condition, malfunction, or request. Precedes work order creation.
PLN (Planned)	PLN (Planifie)	PLN (مخطط)	SAP work order status: order has been created and planned with resources, materials, and scheduling.
REL (Released)	REL (Libere)	REL (صادر)	SAP work order status: order has been released for execution. Materials can be reserved and work can begin.
PCNF (Partially Confirmed)	PCNF (Partiellement confirme)	PCNF (مؤكد جزئياً)	SAP work order status: some operations have been confirmed complete, but work remains.
CNF (Confirmed)	CNF (Confirme)	CNF (مؤكد)	SAP work order status: all operations have been confirmed complete by the executing technician.
TECO (Technically Complete)	TECO (Techniquement termine)	TECO (مكتمل فنياً)	SAP work order status: all technical work is finished. Triggers settlement and cost allocation. No further time or material postings.
CTEC (Commercially Complete)	CTEC (Commercialement termine)	CTEC (مكتمل تجارياً)	SAP work order status: financial settlement is complete. Final archive status. Order is fully closed for both technical and commercial purposes.

Domain 4: Software-Specific / Termes logiciels / مصطلحات البرنامج

English	Francais	العربية	Definition
MCP (Model Context Protocol)	MCP (Protocole de contexte de modele)	بروتوكول سياق النموذج (MCP)	Standardized protocol enabling AI models to interact with external tools and data sources through structured tool definitions and execution contexts.
Agent (AI Agent)	Agent (Agent IA)	وكيل (وكيل الذكاء الاصطناعي)	Autonomous AI component that executes multi-step workflows: field capture classification, planner recommendation, backlog optimization. Operates under human-in-the-loop approval.
Tool Wrapper	Enveloppe d'outil (Wrapper d'outil)	غلاف الاداة	Software layer that exposes deterministic engine functions as callable tools for AI agents via the MCP protocol. Ensures structured input/output contracts.
Engine (Deterministic Engine)	Moteur (Moteur deterministe)	محرك (محرك حتمي)	Core business logic module that produces repeatable, auditable results (e.g., RCM decision tree, criticality calculator, RPN calculator). No AI randomness.
Validator	Valideur	المدقق	Component that checks data integrity against business rules: 72 valid mechanism-cause combinations, task naming conventions (max 72 chars), work package naming (max 40 chars, ALL CAPS).
Processor	Processeur (Traitement)	المعالج	Pipeline component that transforms raw data through multiple stages: field capture text processing, work request structuring, SAP upload package generation.
Hierarchy Node	Noeud de hierarchie	عقدة التسلسل الهرمي	Element in the 6-level plant hierarchy: Plant - Area - System - Equipment - Sub-Assembly - Maintainable Item. Each node has a type, parent reference, and metadata.
Health Score	Score de sante	درجة الصحة	Composite metric (0--100) reflecting overall equipment condition calculated from criticality class, backlog hours, failure mode coverage, and strategy completion.

English	Francais	العربية	Definition
Traffic Light (KPI Indicator)	Feu de signalisation (Indicateur KPI)	إشارة المرور (KPI مؤشر)	Visual KPI status using color codes: Green (on target), Yellow (warning/approaching threshold), Red (off target/critical). Used in executive dashboards.
Backlog Stratification	Stratification de l'arriere	تصنيف التراكم	Analytical breakdown of maintenance backlog by blocking reason, priority level, and equipment criticality to identify bottlenecks and optimize scheduling.
Milestone Gate	Porte de validation (Jalon)	بوابة الانجاز	Approval checkpoint in the maintenance workflow where human review is mandatory before proceeding. Examples: work request validation, schedule approval, SAP upload authorization.
Session State	Etat de session	حالة الجلسة	Streamlit application state persisted across page interactions, storing selected plant, equipment, filters, and workflow progress for each user session.
Equipment Library	Bibliotheque d'equipements	مكتبة المعدات	Reusable catalog of standard equipment types (Pump, Conveyor, Crusher, Mill, etc.) with pre-defined failure modes, maintenance strategies, and component lists. Sources: R8 Library, OEM, Custom.
Component Library	Bibliotheque de composants	مكتبة المكونات	Reusable catalog of standard components (Mechanical, Electrical, Instrumentation, Structural, Hydraulic, Pneumatic) with associated failure mechanisms and maintenance tasks.
Vendor Data Import	Import de donnees fournisseur	استيراد بيانات المورد	Feature for ingesting external data (Excel/CSV/JSON) from equipment manufacturers or SAP extracts. Includes validation against schema rules before database insertion.
FMECA Worksheet	Feuille de travail AMDEC	ورقة عمل AMDEC	Digital form for conducting FMECA analysis through 4 stages: (1) Asset definition, (2) Failure mode entry with RPN scoring, (3) Review, (4) Automated RCM decision assignment.
Approval Status (DRAFT /	Statut d'approbation (BROUILLON / REVISE / APPROUVE)	حالة الموافقة (مسودة / مراجع / موافق عليه)	Three-state workflow ensuring all AI outputs start as DRAFT, pass human REVIEW, and

English	Francais	العربية	Definition
REVIEWED / APPROVED)			require explicit APPROVAL before affecting operations or SAP upload.
SAP Upload Package	Paquet de telechargement SAP	حزمة التحميل الى SAP	Generated data package containing maintenance plans, task lists, and work packages formatted for SAP PM import. Requires human approval before submission (never auto-submitted).
Audit Log	Journal d'audit	سجل التدقيق	Immutable record of all system actions: data changes, approvals, rejections, AI agent decisions, and user interactions. Supports traceability and compliance.

Appendix A: Failure Mechanisms (18) / Mecanismes de defaillance

آليات الفشل

Reference: SRC-09 -- Failure Modes (Mechanism + Cause).xlsx

English	Francais	العربية	Description
Arcs	Arcs electriques	اقواس كهربائية	Electrical arcing between conductors
Blocks	Obstrue / Bouche	انسداد	Flow or passage obstruction
Breaks / Fracture / Separates	Casse / Rupture / Separe	كسر / تصدع / انفصال	Structural separation under load
Corrodes	Corrode	تآكل كيميائي	Chemical or electrochemical degradation
Cracks	Fissure	تشقق	Surface or structural cracking
Degrades	Se degrade	تدهور	General material degradation
Distorts	Se deforme	تشوه	Shape or dimensional change
Drifts	Derive	انحراف	Parameter drift from specification
Expires	Expire	انتهاء الصلاحية	Time-limited component expiration
Immobilised	Immobilise (Bloque / Grippe)	تجمد (انحشار)	Seized, stuck, or frozen condition
Looses Preload	Perd la precharge (Desserrage)	فقدان الشد الاولي	Fastener or connection loosening
Open Circuit	Circuit ouvert	دائرة مفتوحة	Electrical discontinuity
Overheats / Melts	Surchauffe / Fond	سخونة مفرطة / انصهار	Thermal failure exceeding limits
Severs	Coupe (Sectionne)	قطع (بتر)	Complete separation by cutting action
Short Circuits	Court-circuite	دائرة قصيرة	Unintended electrical connection
Thermally Overloads	Surcharge thermique	حمل حراري زائد	Thermal capacity exceeded
Washes Off	Lessivage (Lavage)	غسل (ازالة بالتدفق)	Surface treatment or coating removal by fluid
Wears	S'use (Usure)	تآكل ميكانيكي (بلى)	Gradual material loss through friction

Appendix B: Abbreviation Quick Reference / Reference rapide des

abreviations / مرجع سريع للاختصارات

Abbreviation	English	Francais	العربية
ACR	--	Analyse des Causes Racines	تحليل السبب الجذري
AMDE	--	Analyse des Modes de Defaillance et de leurs Effets	تحليل انماط الفشل وآثارها
AMDEC	--	Analyse des Modes de Defaillance, de leurs Effets et de leur Criticite	تحليل انماط وآثار وحرجية الفشل
CBM	Condition Based Maintenance	Maintenance Basee sur la Condition	الصيانة المبنية على الحالة
CMMS	Computerized Maintenance Management System	GMAO (Gestion de Maintenance Assistee par Ordinateur)	نظام ادارة الصيانة المحوسب
DE	Defect Elimination	Elimination des Defauts	ازالة العيوب
EQUNR	Equipment Number (SAP)	Numero d'equipement (SAP)	(SAP) رقم المعدات
FFI	Failure Finding Interval	Intervalle de Detection de Defaillance	فترة اكتشاف الفشل
FMEA	Failure Mode and Effects Analysis	AMDE	تحليل انماط الفشل وآثارها
FMECA	Failure Mode, Effects and Criticality Analysis	AMDEC	تحليل انماط وآثار وحرجية الفشل
FT	Fixed Time (Maintenance)	Temps Fixe	وقت ثابت (صيانة)
GdC	--	Gestion du Changement (MoC)	ادارة التغيير
IBR	--	Inspection Basee sur le Risque (RBI)	التفتيش المبني على المخاطر
JFC	Jorf Fertilizer Complex	Complexe d'Engrais de Jorf Lasfar	مجمع الاسمدة بالجرف الاصفر
KPI	Key Performance Indicator	Indicateur Cle de Performance (ICP)	مؤشر الاداء الرئيسي
LCC	Life Cycle Cost	Cout du Cycle de Vie (CCV)	تكلفة دورة الحياة
LOTO	Lock Out Tag Out	Consignation / Deconsignation	القفل والتعليق
MBC	--	Maintenance Basee sur la Condition (CBM)	الصيانة المبنية على الحالة

Abbreviation	English	Francais	العربية
MBF	--	Maintenance Basee sur la Fiabilite (RCM)	الصيانة المرتكزة على الموثوقية
MCP	Model Context Protocol	Protocole de Contexte de Modele	بروتوكول سياق النموذج
MoC	Management of Change	Gestion du Changement	ادارة التغيير
MTBF	Mean Time Between Failures	Temps Moyen entre Defaillances	متوسط الوقت بين الاعطال
MTTR	Mean Time to Repair	Temps Moyen de Reparation	متوسط وقت الاصلاح
NPR	--	Nombre de Priorite de Risque (RPN)	رقم اولوية المخاطر
OCR	Optimal Cost Replacement	Remplacement a Cout Optimal (RCO)	الاستبدال بالتكلفة المثلى
OEE	Overall Equipment Effectiveness	Taux de Rendement Global (TRG)	الفعالية الشاملة للمعدات
OT	--	Ordre de Travail (Work Order)	امر العمل
PT	--	Paquet de Travail (Work Package)	حزمة العمل
RBI	Risk-Based Inspection	Inspection Basee sur le Risque	التفتيش المبني على المخاطر
RCA	Root Cause Analysis	Analyse des Causes Racines	تحليل السبب الجذري
RCM	Reliability Centered Maintenance	Maintenance Basee sur la Fiabilite (MBF)	الصيانة المرتكزة على الموثوقية
RPN	Risk Priority Number	Nombre de Priorite de Risque (NPR)	رقم اولوية المخاطر
RTF	Run to Failure	Maintenance Jusqu'a la Defaillance	التشغيل حتى الفشل
TPLNR	Technical Place Number (SAP)	Numero de Poste Technique (SAP)	رقم الموقع الفني (SAP)
TRG	--	Taux de Rendement Global (OEE)	الفعالية الشاملة للمعدات
WO	Work Order	Ordre de Travail (OT)	امر العمل
WP	Work Package	Paquet de Travail (PT)	حزمة العمل

*Generated for the OCP Maintenance AI MVP -- Reliability Centered Maintenance Platform
Office Cherifien des Phosphates -- Jorf Lasfar, Morocco*